UNSW

AUSTRALIA

School of Computer Science and Engineering

Faculty of Engineering

The University of New South Wales

COMP3421 - Project Proposal

by

Ana Yupanqui Ponce De Leon

COMP3421 - Computer Graphics

Bachelor of Engineering in Software Engineering

Submitted: October 2025 Student ID: z5406260

Contents

Introduction....2
1. Comparison with Existing Systems....3
Companion Dynamics....3
Tension and Noise Mechanics....4
World-Building and Storytelling: Cutscenes....5
Inventory Items....6
Enemy Behaviours....8
Combat Mechanic and Weapons....10
2. The Purpose of All Systems....12
2.1. The Overall Purpose of the System....12
2.2 Addressing Problems Identified in Existing Systems....12
Companion Reliability....12
Overuse of Cutscenes and Passive Storytelling....13
Unrealistic inventory....13
Enemy and Combat Balance....13
2.3 Purpose of Individual Module of the System....14
The Safehouse....14
The City....15
The Bodega / Grocery Store....15
3. Different Features of the System....17
3.1 Technical Features....17
3.2 Non-Technical Features....20
4. Novelty of the System....21
5. System Flow....23
5.1 Overall System Flow....23
5.2 Level Flow....24
5.3 Duke State Flow....25
5.4 Zombie AI Flow....26
6. Wireframe/Mock-up of System....27
6.1 Module Overview Map (top-down layout)....27
6.2 Safehouse Interior (top-down layout)....28
6.3 Bodega Interior (top-down layout)....29
6.4 City Segment (Rescue scene)....30
6.5 Inventory....31
6.6 Menu....31
6.7 General Game Play Screens....32
Noise system + Commands + Duke proximity system....32

Throwing Item + Aiming mechanism..... 32
Players perspective - third person.....33
General Night Lighting + Atmosphere..... 33
6.8 Ending screens..... 33
Win Screen..... 34
Lose Screen.....34

Introduction

I intend to develop a third-person, single-level survival game where the player and their dog, Duke, are both trying to survive a zombie apocalypse where noise can mean the difference between life and death. The emotional spin of this game addresses the nightmare scenario many pet owners dread where survival alongside their beloved fluffy best friend, a regular, non-military trained animal, leaves them constantly balancing on the razor's edge of terror and comfort. Additionally, the game incorporates a heart-wretching storyline, noise-based stealth, and heavy tension moments that can have detrimental butterfly effects.

The main aim of this game is for both player and dog to survive in a zombie infested city while scavenging for essential supplies and to safely return to their starting position at the safehouse. Through this journey, the player must manage noise, avoid noise-reacting zombies, whilst keeping Duke alive with the help of limited commands and whilst only carrying a backpack of dwindling supplies. Duke, however, is not your regular passive companion, he is both an asset and a liability. Duke can skillfully sniff out resources and act as an early alarm system by alerting the player to nearby danger, but his “normal”, suburban upbringing comes with youthful disobedience and reactivity to this new unknown world he now finds himself in.

1. Comparison with Existing Systems

Companion Dynamics

Many end-of-world games include some kind of human-centred companion dynamic that holds the foundation of emotional story telling. One of the most notable examples is featured in “The Last of Us”, where the relationship between Joel and Ellie creates a strong emotional hook to players as it depicts the deep human need to protect and connect, which makes them as a pair integral to the story and gameplay.

The main aim of systems like this is relying on the player wanting to save both characters. The companions are usually depicted as essential but fragile, unpredictable and even a little unhelpful. For example, Ellie is a teenage girl who holds the key to the cure but is often a liability for Joel. There are moments where she runs off, gets captured, or acts against Joel’s plans, forcing the player to adapt and take risks. This unreliability creates a tense dynamic where the player feels constant responsibility but never full control, which keeps the emotional stakes high. An extra bonus is that, this kind of companionship makes it adamant to the player that they cannot truly win without trying to save them both. “The Last of Us” is one of many examples that showcase human partners with emotional depth but fewer explore unreliable animal companions where verbal reasoning is limited.

Typically animals, especially dogs, in survival games are more like tools for the player as they are very obedient and often tactically trained. This dehumanises the animal into an object the player can use rather than a living companion central to the player's mental, emotional and physical well-being. Eg. the K9 unit in “Call of Duty Black Ops 2”.

Tension and Noise Mechanics

In horror and thriller games like “Until Dawn” tension and noise mechanics are central as to their success and unpredictability in gameplay. “Until dawn” uses a very effective combination of “don’t move” sequences and Quick Time Events (QTEs) that heighten tension and bring another level of player interaction to the game during high suspense moments like when there are monster encounters.

World-Building and Storytelling: Cutscenes

Another highly effective but time consuming mechanic used by “Until Dawn” is its reliance on cutscenes and collectables for storytelling. While providing context to drive the video game and increase interest through lore, this aspect can take a lot of time to develop and must be well thought out to ensure storyline cohesiveness without distracting from gameplay. Similarly, “The Quarry” performs a near identical job as “Until Dawn” through the use of collectables that have no real purpose other than providing extra insight and lore to the story through cut scenes to increase player immersion. Additionally, both games do a good job in spacing out the placement of these collectables to prevent any gameplay bottlenecks that prevent the storyline from proceeding.

As positive as they may be towards world building, these kinds of features mean that players have the ability to unlock alternate endings or dialogue between characters as an extra bonus. This provides players with more options and also encourages re-play in order to try to unlock as many different endings as possible.

Inventory Items

Survival games typically implement inventory systems to aid their players, but many go too far in how much a character can realistically carry. Adventure games like “Minecraft” let players hold a large number of tools, weapons, and materials, which since set in an unrealistic-fantasy setting makes sense but doesn't necessarily translate over to realistic survival games. Too much inventory can be bad because it removes the tension of having to choose what is essential and what must be left behind. Overall it removes the realism associated with fear of being unprepared which is unauthentic to real-inspired end-of-world scenarios. Realistically, regular people would not be able to carry multiple weapons along with many tools they may need as it would slow them down and possibly get them killed.

In contrast, games like “The Quarry” and “Until Dawn” strip down what characters can carry and focus on the collection of clues, evidence and tarot cards that aid in story telling. The most we see characters be able to carry are things like a gun (which can be held by one character at a time and has very limited ammunition), a flashlight/torch, or items that are either used immediately or lost. This type of unhelpful or non-traditional inventory heightens the fear factor and makes players feel vulnerable. Additionally, it adds importance to the decisions made by the players during QTEs by mirroring closer to how a real person might survive in this scenario and less about collecting resources.

Enemy Behaviours

In “Resident Evil 2” the depiction of zombies aligns with that of the classical zombie trope of slow-moving, persistent monsters that provide balance to the system as while being a tension filled interaction it also gives the player time to run away or fight. Coupled with weapon access, the system drifts away from a stealth and vulnerability lens present in survival-horror gameplay towards action-horror which provides players a greater measure of control.

In contrast, “Until Dawn” creates extremely fast and aggressive enemies that actively hunt the protagonists in adrenaline filled speed chase moments. Paired with a lack of weaponry and a minimal inventory accessible to the protagonists, and a heavy reliance on QTEs, “Until Dawn” forces players to stay in a high-stakes tension filled mindset consistently throughout the game.

Overall, this highlights the difference between games like “Resident Evil” that showcase a balance between combat and persistence based enemies, and “Until Dawn” that strives for unpredictable, adrenaline-filled chase sequences driven by superpowered enemies.

Combat Mechanic and Weapons

Different systems handle combat very differently, and this changes the tone of the whole game. In “Garry’s Mod - Jurassic Park”, players are almost defenseless, armed only with a flashlight and their ability to run. This builds fear but also restricts player agency, since there are fewer tactical choices.

On the other extreme, “Call of Duty Black Ops 2: Zombies” equips players with constant weapon access, making the gameplay closer to an action-shooter. While this is fun and high-energy, the focus shifts away from vulnerability and toward skill in gunplay.

Surprisingly the scariest moments in “COD 2: Zombies” are often when ammo runs dry or when players are trapped in tight spaces like the bus, forced to survive with limited options.

Overall, these contrasting systems between “Garry’s Mod - Jurassic Park” and “COD 2: Zombies” show how weapon access can shape whether a game feels like a survival challenge or a combat simulator.

2. The Purpose of All Systems

2.1. The Overall Purpose of the System

The purpose of this system is to entertain the user through a single-level emotionally immersive survival game that explores the bond between a person and their dog during a zombie apocalypse.

Like many other zombie-survival games, it revolves around the exploration of the player seeing how well they would do if they were put in these post-apocalyptic scenarios. For many individuals in this day and age, end-of-world scenarios are something that constantly looms over, such as the threats posed by global-warming, nuclear war, biological warfare, etc. This aspect of “dooms-day prepping” through survival games profits off the public's fears and worries and through this system it aims to capture the nightmare scenario that every pet owner secretly fears. And unlike many zombie survival games, this system will not emphasise combat, weapons or a larger “saving humanity” plot and instead focuses on the balance between emotional and physical survival to engage players through empathy, attachment and realism. With the overall aim being to ensure the survival of the player and their dog, “Duke”.

2.2 Addressing Problems Identified in Existing Systems

Companion Reliability

Most existing survival games feature companions that are either human and narratively reliable or animal companions that function more like obedient tools. This removes a relatively instinctual unpredictability and sometimes makes relationships feel scripted.

In contrast, this system introduces *Duke*, a loyal yet reactive German Shepherd dog whose regular, suburban upbringing and basic, unperfect obedience training makes him both an asset and a liability. By humanising Duke rather than confining them to breed stereotypes, players must adapt to Duke’s instincts and occasional disobedience. This realism shows that the player’s behaviour and Duke’s behaviour are interdependent and not easily communicable which creates tension during high-stakes moments. This allows players to project their own real experiences of care, frustration, and attachment, ideally creating an outcome that is grounded and emotionally charged.

Overuse of Cutscenes and Passive Storytelling

Games like “Until Dawn” and “The Quarry” rely heavily on cutscenes to deliver lore and emotion. While effective for storytelling, these features are heavily time consuming considering the one-level nature of this system.

Instead, this system will attempt passive storytelling by embedding environmental cues and micro-interactions such as claw marks or blood trails, and ambient design that conveys backstory without explicit dialogue. This minimal *“show-don’t-tell”* approach maintains flow in story telling without the need to create a deep lore for the system.

Additionally, through the use of QTEs, this system will maintain tension and provide a fast paced story telling experience that ideally keeps players engaged.

Unrealistic inventory

Traditional survival games often let players carry excessive supplies which undermines tension that comes with being unprepared or under-equipted. Instead, this system will implement a realistic inventory that's limited to a small backpack that only holds a few items such as some food, water, and a few tools. Players must make difficult decisions about what to take or leave behind as it will directly affect their chances of survival. Through this, the aim is to restore authenticity and ensure the fear of scarcity remains constant.

Enemy and Combat Balance

Many games either feature slow, predictable enemies or overwhelming chase sequences. This system combines both to sustain pressure without losing realism. Through classical-style enemies, the system’s zombies react primarily to noise rather than vision, meaning tension depends on player stealth rather than combat skill. Additionally, the player’s access to no weaponry makes every encounter a risk-versus-reward decision. Causing a loud distraction might save Duke but may also attract new threats which aligns with the core theme that *every sound has consequences*.

2.3 Purpose of Individual Module of the System

As a single-level system, the module contains three interconnected environments; the safehouse, the city area, and the grocery store (bodega).

The overall purpose of this module is to immerse players in a tense, cinematic survival experience where the primary goal is to scavenge food and return to the safehouse alongside their dog companion, *Duke*. Although these areas exist within the same map and singular module, each area/environment has its own mechanics and challenges to create engagement and tension while maintaining a consistent and cohesive post-apocalyptic New York-inspired atmosphere and aesthetic.

Additionally, the gradual transition from daylight to night throughout the module reflects the growing danger and urgency of the player's mission as zombie numbers increase and player vision lessens with nightfall.

The Safehouse

The module begins with the player and Duke inside a small, run-down safehouse which also serves as the player's spawn point and initial safe zone.

Design wise, it is a dimly-lit, barebones, dirty, claustrophobic, one-floor apartment that feels *just* safe enough to sleep in. As for safety reasons, all possible entrances like windows are boarded up and doors are heavily locked. Here players can find an old mattress, an empty backpack, an old food bowl for Duke, and most alarmingly several empty cans of food and water bottles. The minimal items available immediately communicate scarcity and signaling to players that they are already in an extremely vulnerable state.

From a gameplay perspective, the safehouse is designed to be simple and rundown with minimal game mechanics and interactable items to drive players to venture into the city. Players can decide to stay in the safehouse and risk starvation for themselves and Duke, or they can take a risk and venture into the city and hope for a safe return.

Additionally, to reinforce Duke's importance, the safehouse key is stored in a pocket within Duke's tactical harness to keep it safe, thus making his safety essential for both progression and survival. Without Duke's safe return, players will not be able to re-enter the safehouse which reinforces the physical dependency between player and companion from the start.

The City

Directly outside the safehouse, the player enters a city area that continues the post-apocalyptic theme and expands the environment available to the player and simultaneously raises the safety stakes. The city is inspired by a distorted version of a rundown, dark, and eerily quiet New York city with very few working streetlamps, flickering billboards, abandoned buildings and scattered debris. As previously mentioned, the lighting shifts from a well-lit day to a soft dusk as players progress which mirrors an increasing difficulty and tension. Coupled with sound immersion such as footsteps that echo in the deserted streets, distant moans of roaming zombies, and the occasional growl or whimper from Duke when zombies are near, the city is the main obstacle in the way of the player's successful supply run.

Gameplay-wise, the city functions as a transitional survival zone with a relatively simple and small explorable area of a few city blocks divided by a main road and splitting alleyways. This gives it a maze-like feel that separates the safehouse and grocery store as players do not have access to a map. The city itself is not very interactable but does offer a few areas for the player to hide from roaming zombies in alleyways and other objects such as dumpsters, abandoned cars, or debris.

The alleyways between blocks are very short and lead to dead ends and serve mainly to allow players the opportunity to hide and stay quiet whilst looking for the grocery store. The buildings themselves are not explorable to the player as that would distract from the straight forward design of the game by allowing players to take an exploratory route rather than focusing on pure survival. This way, players are forced to zig zag through alleys to reach their goal destination (safehouse or grocery store). The city is where the player's interactions with Duke through commands like "stay", "quiet", and "follow" become pivotal as any errors can worsen high-risk situations. Overall, this forces players to remain vigilant and pick their interactions carefully as the city is built to demonstrate the sound and motion driven systems that drive zombie behaviour.

The Bodega / Grocery Store

At the opposite end of the city from the safehouse, the players can find a small corner grocery store modelled after a New York-style bodega, complete with its own bodega cat that freely comes and goes. Keeping in mind a post-apocalyptic New York, the bodega interior consists of a small one floor shop full of half-empty shelves, powered-down fridges, thick dust coating surfaces and an unserviced register area. Inside you can find scattered general store supplies like canned food, bottled water, cleaning supplies and pet food.

The grocery store area serves as a midway and climax point in the module. Here players have the opportunity to complete one of their primary goals (to re-supply food and water). Additionally, the bodega is more interactable than the city and safehouse and can be considered a semi-safe zone for the player and Duke where they can roam between the aisles to search for supplies, or venture behind the register area for quick cover. However, the bodega's broken, open doors make it easy for anyone or anything to enter and it's pivotal for the player and Duke to remain quiet as zombies lurk not too far from the entrance.

Here a critical narrative event occurs where Duke runs off after the bodega cat after it makes an impromptu appearance. This event forces the player to test all the skills they've learnt like stealth, distraction use, and Duke's commands to rescue him before zombies get him and so both the player and Duke can return safely to the safehouse.

3. Different Features of the System

3.1 Technical Features

Player Controls and Interactions

- Player is controlled through keyboard and mouse inputs.
- **Movement**: Walk, crouch, sprint.
- **Interactions**: Open door, pick-up and throw items, and give Duke commands.
- Interaction prompts and instructions will appear only when in close range to interactable objects.

Companion Command System

- The player can issue basic verbal commands with Duke: ***Stay, Quiet, Follow***.
  - Indicated through helper keys.
  - Press “J” key for “follow”
  - Press “K” key for “stay”
  - Press “L” key for “quiet”
- Commands have limited timed duration before Duke reverts to autonomous behaviour.
  - **Command timer bar** visually indicates the remaining obedience time.
- Obedience and reliability decrease during high-stress conditions like when enemies are near and fully don’t work when Duke is distant.
  - **Stress bar** visually indicates Duke’s stress levels at any time.
  - High stress levels are more likely to end with Duke barking, breaking a command, or running away.
- Reinforces Duke’s realistic behaviour as he may disobey or react unpredictably under stress.

Quick Time Events (QTEs)

- Used during key tension moments and have a default option if the player cannot respond in time. Examples include:
  - When Duke chases the bodega cat
    - **Choise #1:** chase him immediately.
    - **Choice #2:** stay hidden until zombies pass.
  - When Duke is cornered by zombies
    - **Choice #1:** create a distraction.
    - **Choice #2:** attempt a command.
- QTEs add urgency and immediate consequences to player decisions.

Inventory and Throwable Items System

- Player carries a small backpack with limited slots for food, water, and throwable items.
    - Throwable items include cans of food (non-empty and empty).
    - Backpack is old and worn and with small holes.
- Inventory items can be dropped or thrown to create sound and motion based distractions to attract or deflect zombies.
- Aiming/Projectile path trajectory feature available but not super accurate
- This feature encourages resource management and tactical use of supplies.

Physics simulations

- Basic gravity and collision physics will be implemented. Will aid with:
    - Thrown items
    - Environmental debris
    - Ragdoll effect (when character or Duke dies)

Sound and Motion-Based Enemy System

- Zombies react to **movement and sound** rather than relying solely on vision.
- Both player and Duke can attract attention if noisy or moving too fast.
    - Player and Duke make noise through footsteps, barking, giving Duke commands or dropping/throwing items.
    - **Noise meter** provides visual feedback to help player stay quiet
- This requires players to remain still and/or silent in order to avoid detection.

Lighting and Time-of-Day System

- Gradual and automatic shift from daylight to night.
- Environmental lighting: Flickering streetlamps, signs and moonlight used to illuminate the city during night time.

Objective and Message System

- Small message bubble that appears during gameplay to communicate objectives, story cues, or environmental warnings.
    - *"Find Duke before it's too late!"*
    - *"Stay quiet, there are zombies nearby".*
    - *"Yay! You found the bodega!"*
    - *"Supplies collected. Return to safehouse."*

Completion conditions

- Win: Collect supplies and get back to the safehouse safely with Duke.

- Lose: Player and/or Duke dies, or returns to the safehouse without supplies.

Game mechanics

- Standard save game system
- Duke mostly follows the player but not always closely and can drift away exploring the surroundings.
- Noise detection and stealth system
    - Noise meter
    - Duke stress level bar
    - Duke command timer bar
- Key to the safehouse stored in Duke's harness.
- Environmental interaction feedback
    - Surfaces and debris produce different noise levels when touched or hit.
- Respawn system is always at the safehouse and at the beginning of the game.
- Message system to signal new goals, player progress and player's thoughts.

Environment Specific Features

- Safehouse
    - Features limited lighting and interaction objects (used supplies).
    - Has lock on door connecting to the outside.
        - Player always chooses to keep the key in Duke's harness.
        - Introduces Duke's basic commands (stay, follow, quiet).
        - Introduces player's movement and interaction controls (walk, crouch, pick up interactable items).
- Bodega:
    - Supplies are only found in the bodega and player is able to interact.
    - Main inventory and resource management section.
    - Physics-based interaction section as items on shelves and floor can be bumped into and create noise.
- City
    - Implements day-night system
        - As night comes, the working street lamp and lighting turn on
    - Exposes the player to a more diverse noise system through objects that create different levels of sounds when bumped into or interacted with (car, cans, fences, trash cans).
    - Duke's stress system becomes active and fluctuates more in this environment.
    - Quick Time Events start.

3.2 Non-Technical Features

Visual Theme and Environment Design

- Distorted, post-apocalyptic New York suburb setting.
- Features include boarded-up buildings, flickering signs, abandoned vehicles, debris, and littered streets.
- Evidence of carnage present, such as smeared blood and dead body parts

Storyline and Narrative

- Linear, story-driven design focused on joint survival divided in four acts.
- **Act 1: The safehouse**
  - The player wakes in a small, rundown safehouse with Duke beside them and minimal supplies. Scarcity forces the player to leave the safehouse to find food and water in the nearby city.
  - Introduces basic movement, stealth, and command mechanics.
- **Act 2: The bodega**
  - On the journey through the city, the player encounters scattered zombies, testing Duke's obedience and the player's stealth skills.
  - After re-supplying, the bodega cat appears and Duke bolts after it into the streets. His barks attracting nearby zombies.
  - The player must choose to either chase Duke immediately or hide until the zombies pass, triggering a new objective to follow Duke's barks through the city.
- **Act 3: Climax**
  - The player finds Duke trapped and surrounded by zombies.
  - Quick Time Event occurs where the player must choose how to act:
    - Create a distraction, or use a dog command.
  - Player decisions directly affect the survival outcome.
- **Act 4: Resolution**
  - If successful**,** the player reunites with Duke and both return to the safehouse.
  - If unsuccessful, the player and/or Duke don't survive, thus ending in failure.

Audio and Ambient Design

- Minimal music and focus on environmental sounds (footsteps, zombie groans, Duke's breathing, barks and whines).
- Sound cues (Duke growling) used to warn players and signal zombie proximity.

Character, Companion and Enemy Design

- Player**:** Young-adult survivor, under-equipped and vulnerable. Carries a backpack.
- Duke**:** Loyal but imperfect companion; wearing a tactical dog harness.
- Zombies**:** Classical design. Slow, rely on sound and motion for detection.

4. Novelty of the System

The system's novelty stems from combining familiar, survival–horror mechanics with a realistic, emotionally charged dog–companion dynamic that makes both gameplay and narrative outcomes dependent on the player's relationship with Duke. While noise- and motion-based AI is common in survival games, applying these mechanics to both player and animal companion introduces shared vulnerability and tension that is rarely explored.

Addressing Shortcomings in Existing Systems

- Reliable pet companions in survival games
  - In most survival games you typically find animal companions who seem perfectly trained and adjusted to the post-apocalyptic world they find themselves in. Often serving as obedient, utility based tools that can be sacrificed if needed for the sake of the main character in order to win.
  - The obedient military dog stereotype is further expected as Duke is presented as a German Shepherd in a tactical harness. This subverts the game by instead showing Duke's realistic disobedience and stress-driven behaviours, forcing players to treat him as a living creature rather than a controllable asset.
  - Additionally, Duke is a necessary asset towards joint survival and necessary in order to win the game by getting safely back to the safehouse.
- Enemy Sound and Motion mechanics
  - By making both player and companion subject to sound and motion detection, this system closes that realism gap and raises the strategic stakes.
  - This makes stealth a shared responsibility as the player must manage not only their own actions but Duke's imperfect behaviour as well.

Integrated Command Tension Mechanic

- Commands like *Stay*, *Quiet*, and *Follow* have timed durations and emotional reliability tied to Duke's stress level.
- Duke's stress levels shorten command durations and increase likeliness for disobedience under pressure and increase his likeliness to bark, forcing players to adapt in real time.
- This adds realism and unpredictability without the need for complex dialogue trees or combat systems.
- This is also unique because in games with an animal companion, the player does not usually have so much interaction with the dog or is so involved in its survival.
- Here the player is directly tied to keeping Duke calm.

Story

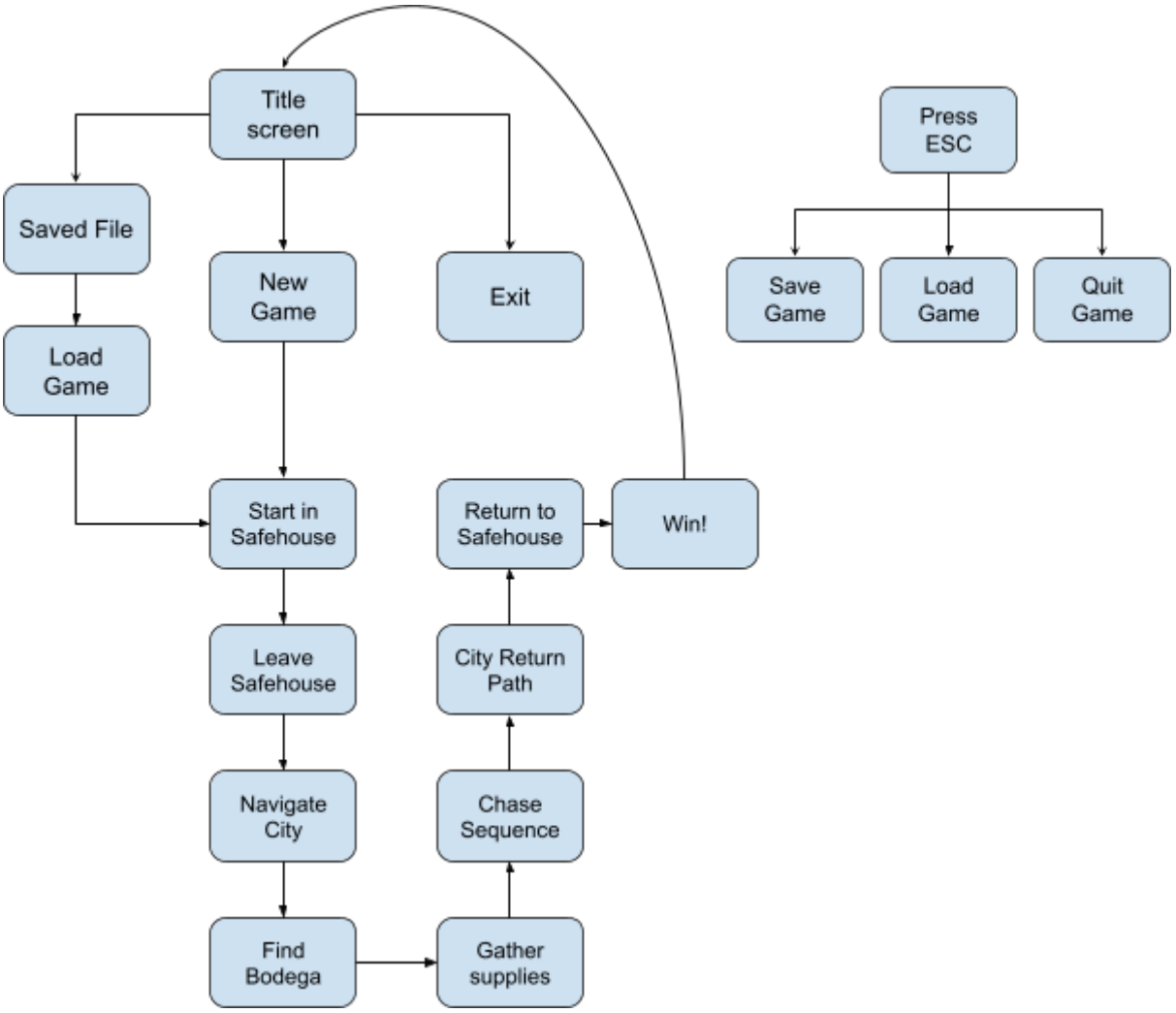
- There have been games that focus solely on player survival and others just on animal survival (Stray's cat protagonist) but I don't think there is one that focuses on the survival of both.

Grounded Survival Realism

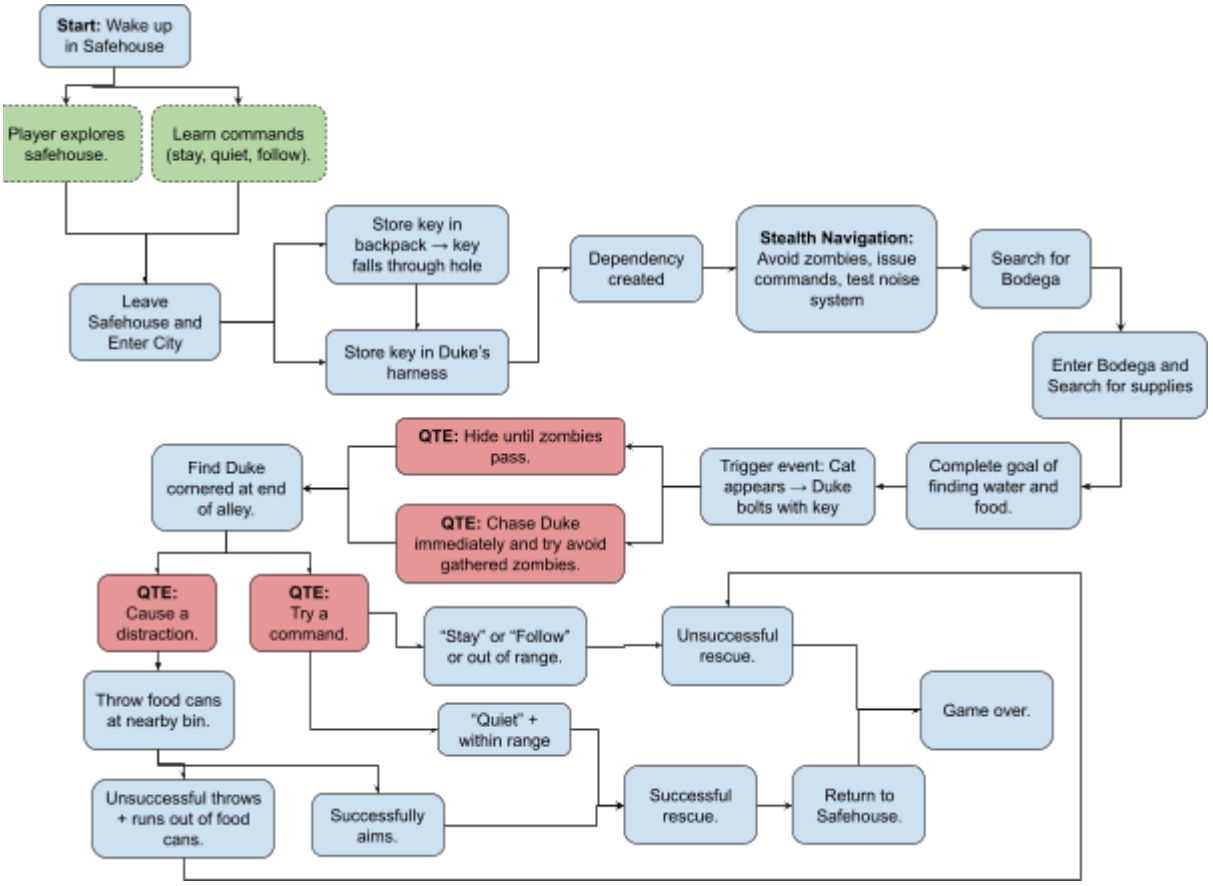
- The player carries a limited inventory, emphasising scarcity and realism over power fantasy.
- Choices like where to store the safehouse key (in Duke's harness or the player's bag) create subtle but meaningful narrative.

5. System Flow

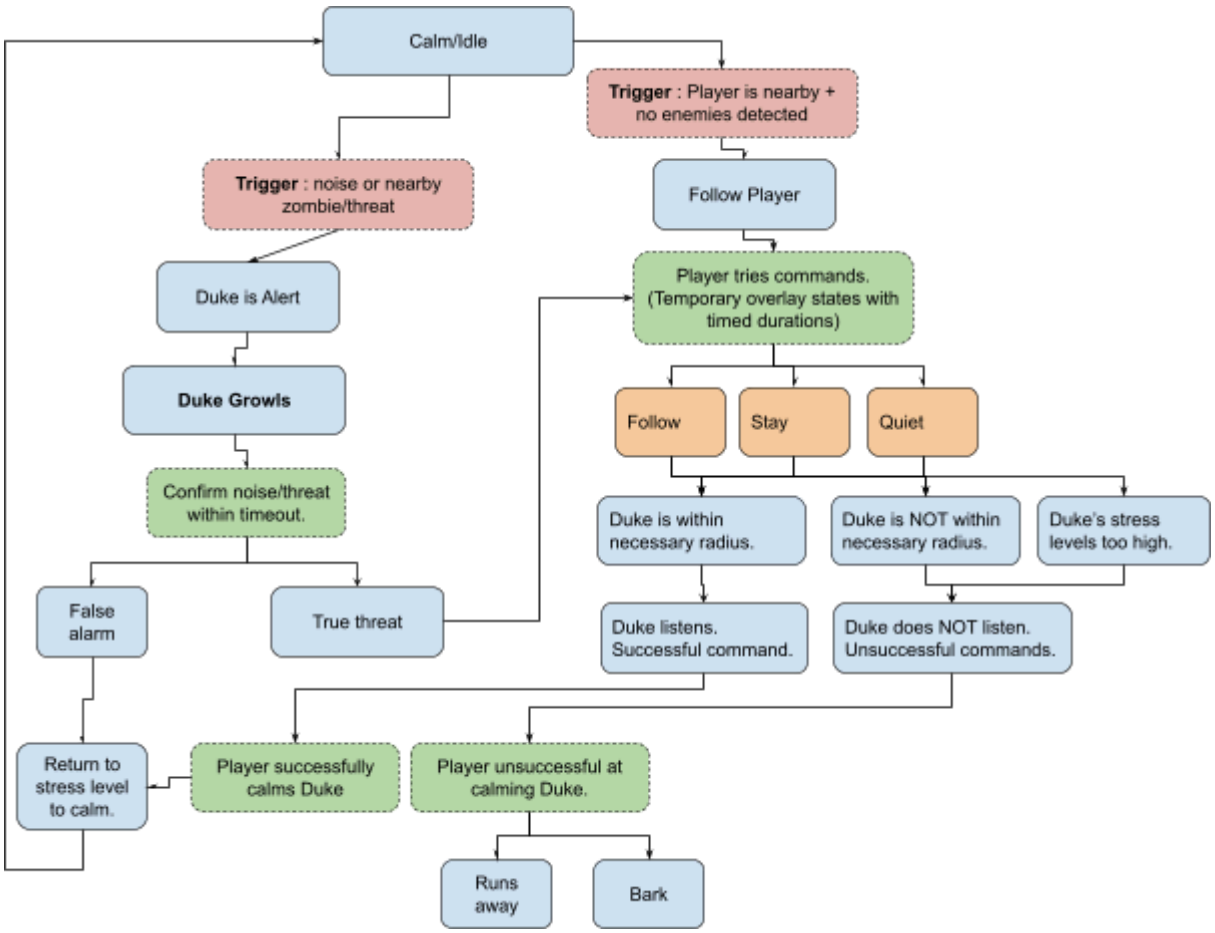
5.1 Overall System Flow



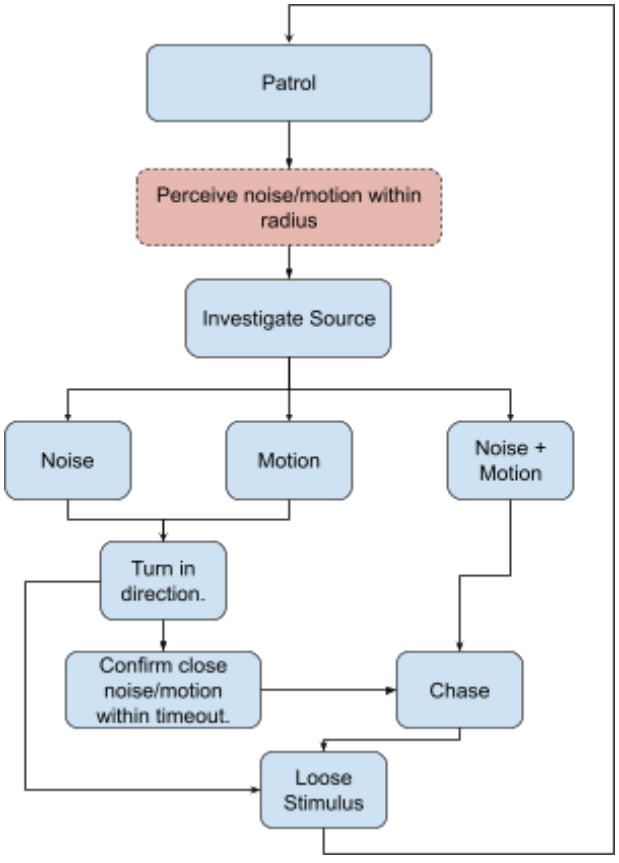
5.2 Level Flow



5.3 Duke State Flow



5.4 Zombie AI Flow



6. Wireframe/Mock-up of System

For this section of the report, I decided to wireframe the key sections of the system that bring technical and story importance towards the system and that would benefit from visualisations to further explain the desired goals and aims.

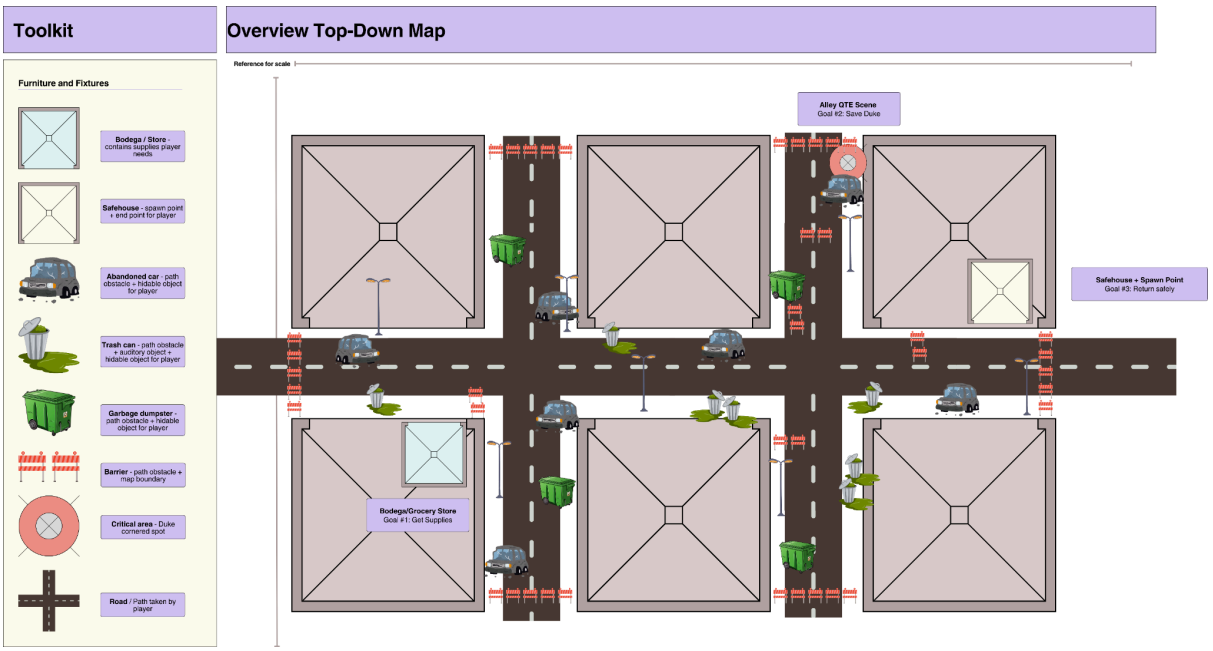
This section expands on the overview of the system map, the safehouse interior, the grocery store (bodega) interior, and lastly, the city segment that will bring the final conflict point in the game.

6.1 Module Overview Map (top-down layout)

Here, an overview of the game environment is shown. The total “explorable” world for this system itself is contained. Due to the single-level nature of this game, it was decided that this “small” world was necessary in order to properly accomplish the system successfully within the given time frame.

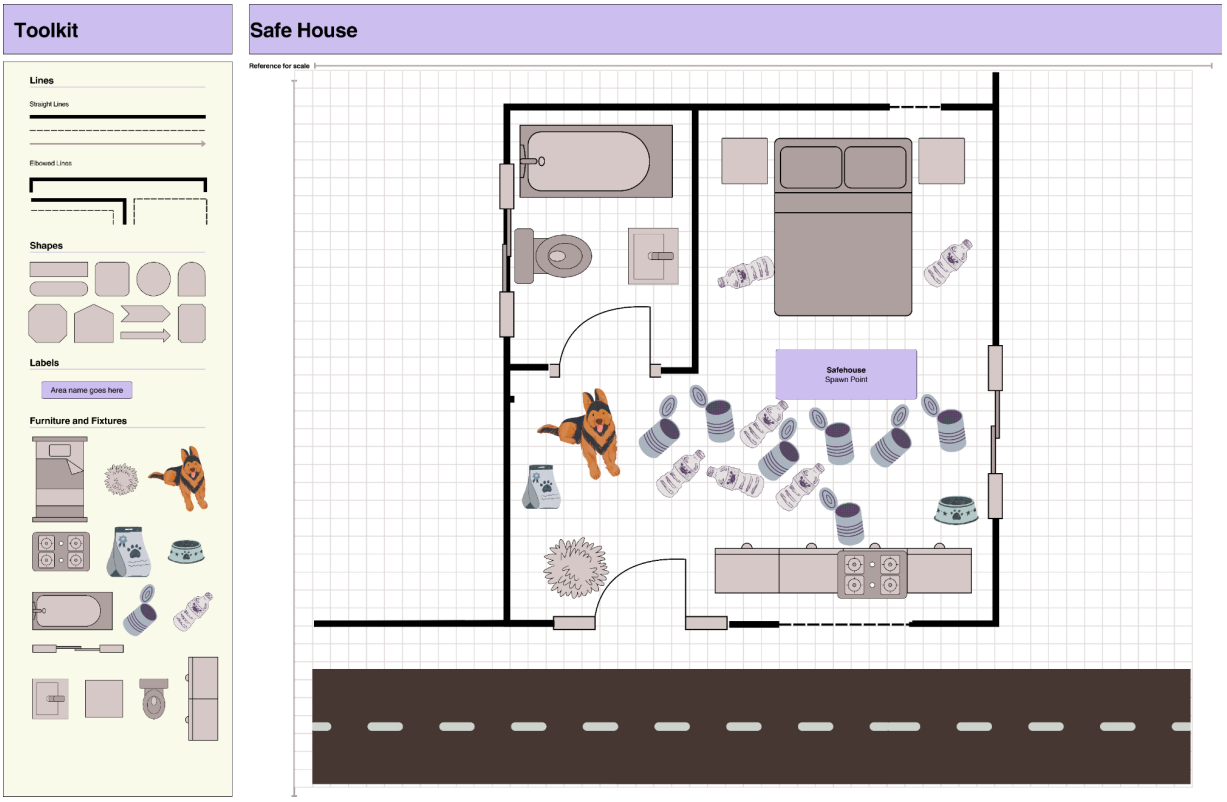
The world is composed of 6x2 blocks of city buildings. With this design, the world is large enough to have several alleyways available for the player to explore whilst looking for the supplies whilst maintaining the straight forward narrative. Various obstacle items can be seen crowding the streets. Some of the items are able to be used by the player to hide, and other items serve as auditory objects the player can interact with.

The 3 key features of the map are the *Safehouse* (far right of map), the *Bodega / Grocery store* (far left), and the *conflict story point* (middle-top-right of the map). More information about the items in the map can be found in the “Toolkit” section below.



6.2 Safehouse Interior (top-down layout)

The design of the safehouse is simple and was done this way as the player is not meant to stay within this environment for too long besides the initial spawning and as the goal destination where the game ends.

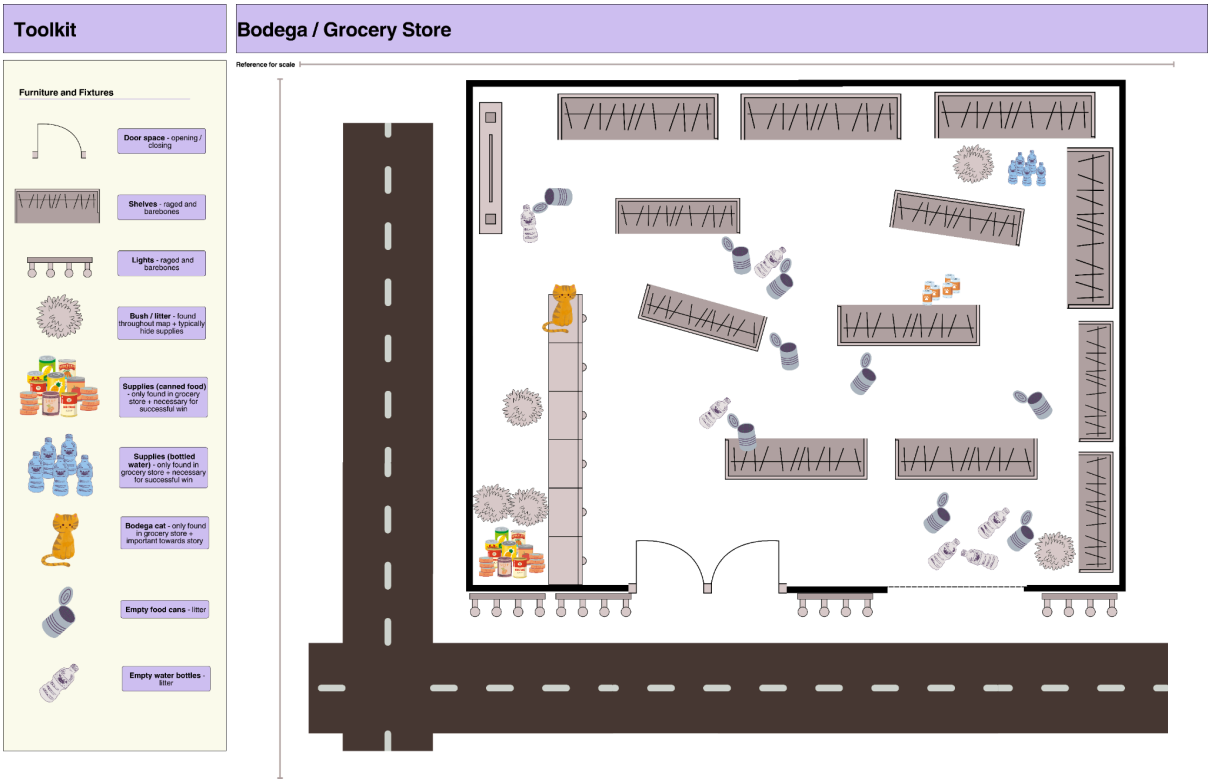


6.3 Bodega Interior (top-down layout)

The design of this environment is also meant to be simple but slightly more obstructive as the player has to try to find the supplies within this “over sacked” grocery store. It can be seen littered with empty food cans and water bottles. In the proper implementation, there will be shelves that are turned over and fallen to further illustrate the dire situation the characters are in. The supplies the player needs will also be semi-hidden by the *bush* items. Perhaps the most important part of this environment aside from the supplies the character has to re-fill, is the *bodega cat* which will trigger the last challenge of the game where the player has to chase after Duke and save him from zombies. The bodega cat is not planned to appear until the player finds the last of the supplies (either water or canned food).

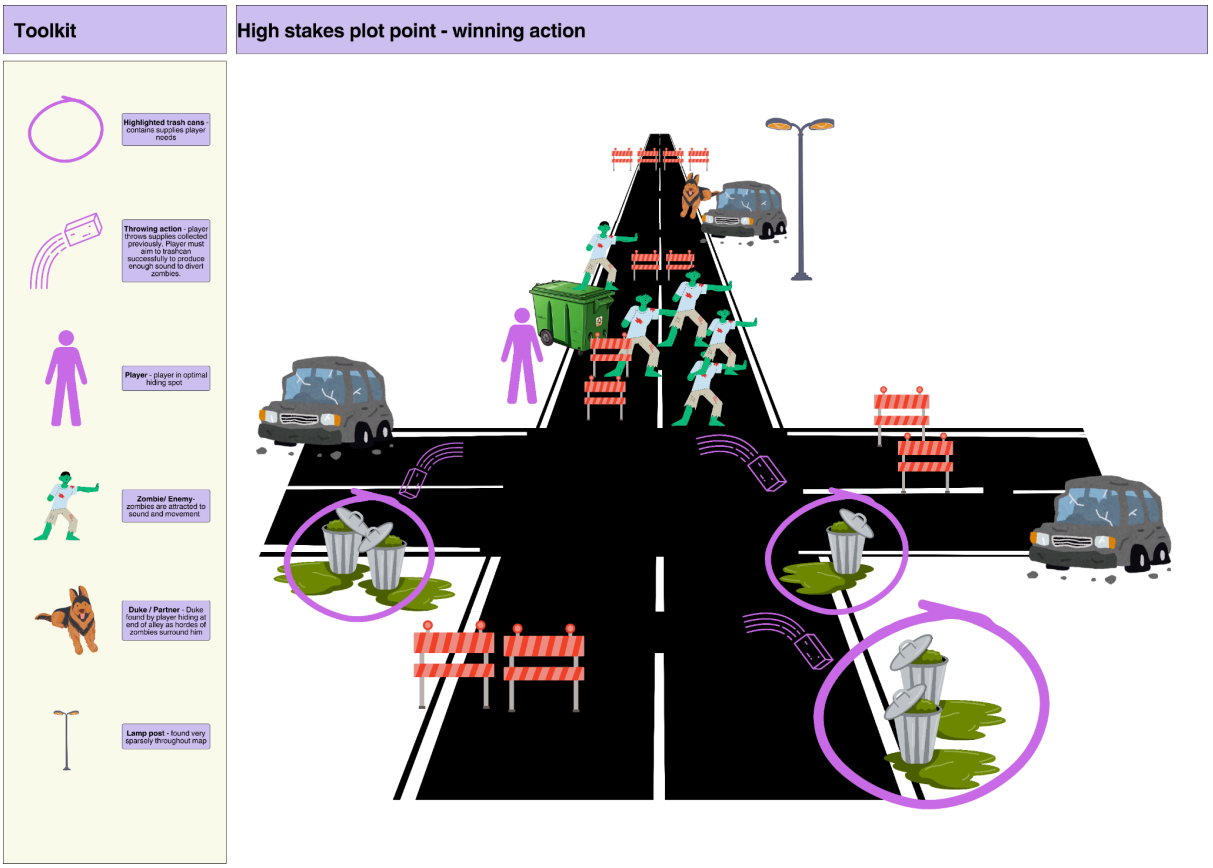
Additionally, the counter section and shelves serve as hiding objects for the player as there will be zombies patrolling outside the store.

Please check “Toolkit” below for more information about the items in this map.



6.4 City Segment (Rescue scene)

This mock-up depicts the last high-tension conflict section of the game where Duke is found hiding at the end of an alleyway surrounded by wandering zombies. Here the player (purple) is going to be faced with a QTE that will decide between the success of the run or will end in impending doom. Below, the winning action is depicted where the player has the opportunity to drive zombies away from Duke by *ideally* successfully aiming and throwing as few canned supplies at any of the circled bins. Throwing supplied at any other object will fail in creating enough noise to distract zombies.



6.5 Inventory

The player will have the option to either “drop” or “throw” items in their bag “inventory”. The inventory itself is very limited to mirror the real life limits a backpack has. The picked items have a highlighted red outline.

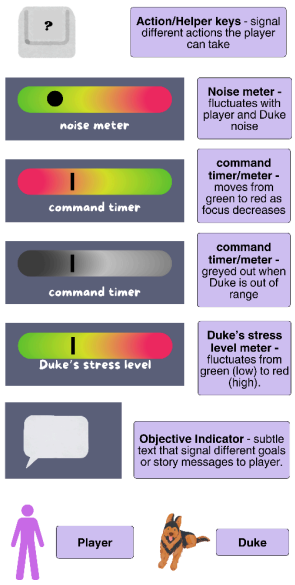
6.6 Menu

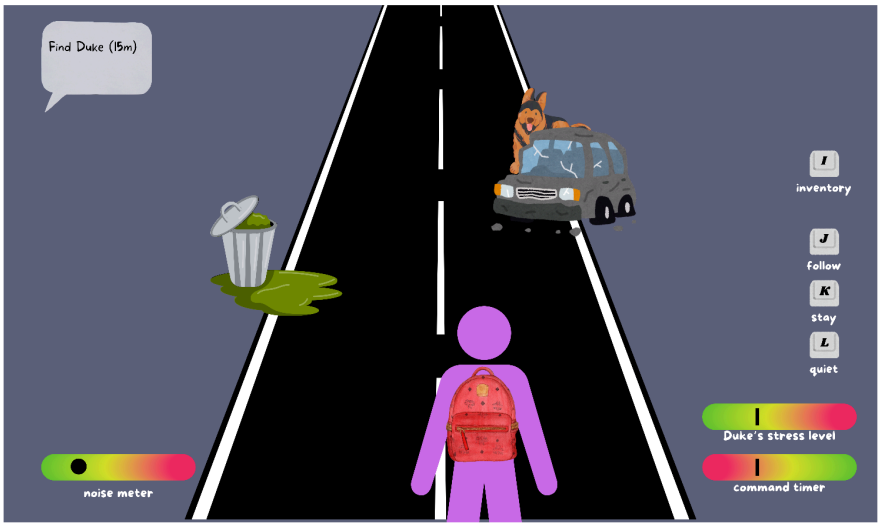
6.7 General Game Play Screens

Noise system + Commands + Duke proximity system

The mock-up below showcases the meter's available to the player. The action keys below are a sample for the mock-up as the end action keys have not yet been fully picked out.

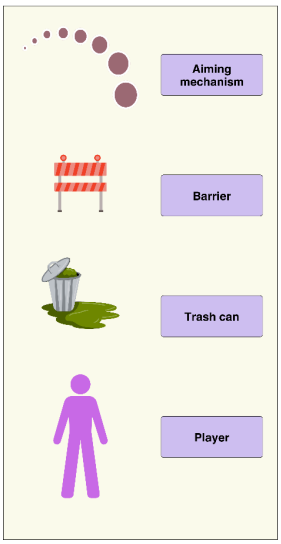
Toolkit	UI Mock-up

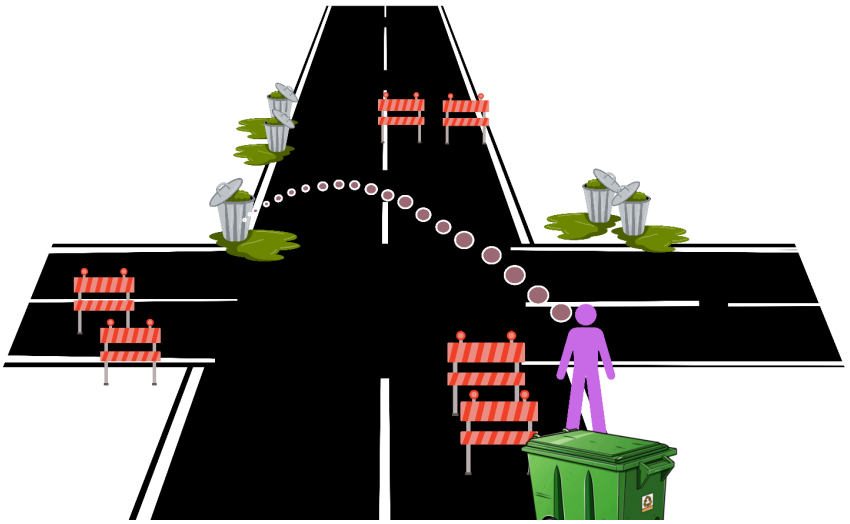




Throwing Item + Aiming mechanism

Toolkit	Throwing action

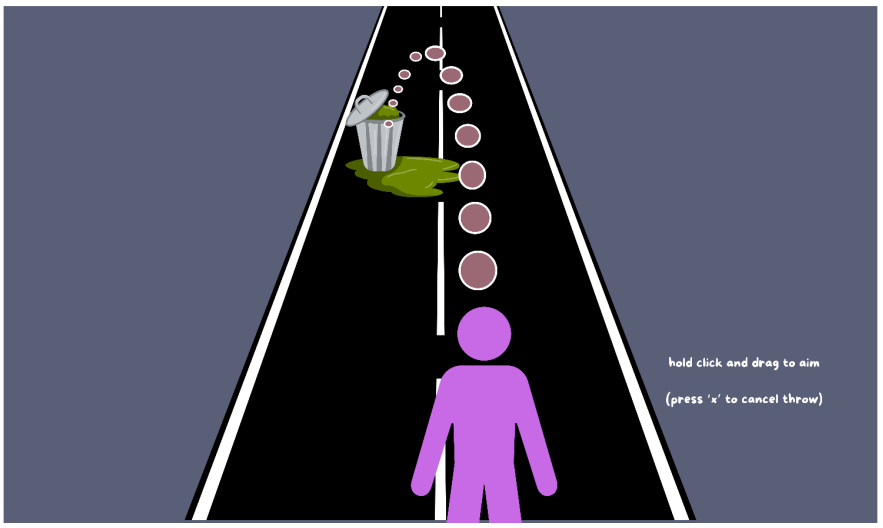




Players perspective - third person

Toolkit

Throwing action (players POV) - aiming mechanism



General Night Lighting + Atmosphere

6.8 Ending screens

Both ending screens are very similar, with the biggest differences being the tint over the background. The background image of both ending screens is not fully cemented and within this mock-up is just a stock sunset picture. The main purpose of this mock-up is to illustrate the options available to the player after they finish playing the game.

Both screens will have the options to try the game again or to exit to the main menu. The option of either will be highlighted with a thicker red text and is additionally accompanied by a red arrow to further highlight the focused option.

Win Screen

The “win” screen will have a more vibrant tint that's similar to the sunset and represents winning to live another day.

Lose Screen

The “lose” screen has a white and black or grey tint that evokes old-cinema aesthetic to represent the death or loss of the game.